



UGANDA INSTITUTE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY

END OF SEMESTER II EXAMINATIONS

DEPARTMENT : ICT

YEAR OF STUDY: : ONE

SEMESTER : TWO

PROGRAM : DIPLOMA IN INFORMATION TECHNOLOGY FOR BUSINESS

: DIPLOMA IN COMPUTER SCIENCE

COURSE : DATABASE SYSTEMS DESIGN I

COURSE CODE : CSC 1203

DATE : SUNDAY, JUNE 30, 2024

TIME : 09:00 AM – 12:00 NOON

DURATION : 3 Hours

INSTRUCTIONS

- ❖ The paper consists of two Sections A and B
- ❖ Answer **ALL** questions in Section A and **ANY THREE** questions in Section B
- ❖ Credit will be given to relevant examples and illustrations
- ❖ Don Not Write anything on this question paper

SECTION A [40 MARKS]:

1. Which of the following is not a type of database?
a. Hierarchical b. Network c. Distributed d. Decentralized
2. What is DBMS?
a. DBMS is a collection of queries b. DBMS is a high-level language
c. DBMS is a programming language d. DBMS stores, modifies and retrieves data
3. Which of the following is a feature of the database?
a. No-backup for the data stored b. User interface provided
c. Lack of Authentication d. Store data in multiple locations
4. Which of the following is a component of the DBMS
a. Data b. Data Languages c. Data Manager d. All of the above
5. What does a RDBMS consist of
a. Collection of Records b. Collection of Keys c. Collection of Tables
d. Collection of Fields
6. The ability to query data, as well as insert, delete, and alter tuples, is offered by _____
a. TCL (Transaction Control Language) b. DCL (Data Control Language)
c. DDL (Data Definition Language) d. DML (Data Manipulation Language)
7. _____ is a set of one or more attributes taken collectively to uniquely identify a record
a. Primary Key b. Foreign key c. Super key d. Candidate key
8. Which command is used to remove a relation from a database?
a. Drop table b. Delete c. Purge d. Remove
9. ____ indicates the maximum number of entities that can be involved in a relationship
a. Greater Entity Count b. Minimum cardinality c. Maximum cardinality
d. ERD
10. What does a foreign key combined with a primary key create?
a. Network model between the tables that connect them
b. Parent-Child relationship between the tables that connects them
c. One to many relationship between the tables that connects them
d. All of the mentioned
11. The oldest DB model is _____
a. Network b. Physical c. Hierarchical d. Relational

12. Which one of the following given statements possibly contains an error?
 - a. select * from emp where empid = 10003;
 - b. select empid from emp where empid = 10006;
 - c. select empid from emp;
 - d. select empid where empid = 1009 and Lastname = 'GELLER';
13. What do you mean by one-to-many relationships?
 - a. One class may have many teachers
 - b. One teacher can have many classes
 - c. Many classes may have many teachers
 - d. Many teachers may have many classes
14. Which one of the following refers to the copies of the same data (or information) occupying the memory space at multiple places.
 - a. Data Repository
 - b. Data Inconsistency
 - c. Data Mining
 - d. Data Redundancy
15. Which of the following refers to the level of data abstraction that describes exactly how the data is actually stored?
 - a. Conceptual Level
 - b. Physical Level
 - c. File Level
 - d. Logical Level
16. To which of the following the term "DBA" referred?
 - a. Data Bank Administrator
 - b. Database Administrator
 - c. Data Administrator
 - d. Data-Based Administrator
17. Which of the following commands is a type of Data Definition language command?
 - a. Create
 - b. Update
 - c. Delete
 - d. Merge
18. In the relational table, which of the following can also be represented by the term "attribute"?
 - a. Entity
 - b. Row
 - c. Column
 - d. Both B & C
19. Which one of the following refers to the customised view of the database content?
 - a. Conceptual view
 - b. Physical view
 - c. Internal view
 - d. External view
20. A computer security protocol for logging-in can be considered as the example of the _____ component of an information system.
 - a. Data
 - b. Software
 - c. Procedure
 - d. Hardware

SECTION B (60 marks)

Question One

- a) You have been approached by a group of first year student of UICT to share with them your knowledge about the Database concepts below. Use your knowledge acquired to explain the following to that group of students

(2 marks @)

- i. Database
- ii. Database Management System
- iii. Data Integrity
- iv. Data Security
- v. Data Redundancy

- b) As an expert of database systems design, advise UICT Management with five (5) reasons why you would discourage the use of File based approach to store and manage the Institute's data. (10 marks)

Question Two

- a) Use a diagram to explain the five (5) components of a DBMS environment. (10 marks)
- b) The ANSI SPARC is a three-level data base architecture. Briefly describe these three levels and what they represent. (06 marks)
- c) In Database systems design, Data Independence is very critical. As a database designer, you know Data Independence as a concept that explains hoe higher levels of the database model are designed to be unaffected by changes to the lower levels (internal and physical). Please explain the two types of data Independence (4 marks)

Question Three

- a) With illustrations, explain the difference between entity relationship model and relational model. (07 Marks)
- b) You have been hired to design a database for a supermarket. The database will hold information regarding customers, orders, order-lines and products. You are given the following conditions to implement in your design:
- Each customer can place one or more order but one order can only be ordered by one customer
 - Each order contains one or more order- lines but one order-line is contained in only and only one order.
 - Every product has one or more order-lines but each order-line can contain one and only one product.

Required:

- (i) Identify the entities and the relationships between the entities identified. (07 Marks)
- (ii) Draw an entity relationship diagram for each of the conditions listed above. (06 Marks)

Question Four

- a) Given your knowledge of Database design, explain the function of SQL in database design and Implementation. (01 Mark)
- b) Give any two Examples of the following:
- i. DDL SQL Commands (2Marks)
 - ii. DML SQL Commands (2 Marks)
- c) In SQL some key words are used as DML Statements to interact with the database. Explain what the statements below do
- i. SELECT (2 Marks)
 - ii. UPDATE (2 Marks)
 - iii. INSERT (2 Marks)
- d) Using the write SQL Commands, please perform the following functions as a Database Administrator
- i. Create a table named "Products" with at least 4 columns with different data types (03 Marks)
 - ii. Using the table above, insert values in the same table (03 Marks)
 - iii. Using a simple query command, please display all recorded data in the table above (03 Marks)

Question Five

Given table **STUDENT** below:

RegNo	Names	Age	Gender
2011/ITB/001	Jack	24	M
2011/ITB/002	Fred	18	M
2011/ITB/003	Allen	22	F
2011/ITB/004	Juliet	24	F
2011/ITB/005	Matthias	25	M
2011/ITB/006	Sarah	16	F

- i) Write an SQL statement to create the above table, assign regNo to be the primary key of the table. (06 marks)
- ii) Write an SQL statement that inserts user data as shown above. (04 marks)
- iii) Write an SQL statement that will display all the information in the table above. (02 marks)
- iv) Write an SQL statement that will display only male students. (02 marks)
- v) Write an SQL statement that will display only students whose names begin with letter J. (03 marks)
- vi) Write an SQL statement that will display students whose age ranges from 17 to 23. (03 marks)

*****GOOD LUCK*****